

HAT-P-55b

R-0018

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-55_260530 · created 2026-07-08T08:33:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461190.882 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.047
Transit depth (Rp/Rs)^2	2.33 +/- 1.43 [%]
Orbital Inclination (inc)	89.05 +/- 2.37
Airmass coefficient 1 (a1)	1.55 +/- 1.02
Airmass coefficient 2 (a2)	-0.33 +/- 1.52
Scatter in the residuals of the lightcurve fit is	66.87 %
Best Comparison Star	None
Optimal Aperture	2.7
Optimal Annulus	18.48
Transit Duration (day)	0.134 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.047 vs 0.1202 (deviation 0.7σ)
Duration fit vs published	0.134 d vs 0.1225 d (deviation 0.77σ)
Mid-time O–C	26.0 min
Depth SNR	3.3
Residual scatter	66.87 %

Light curve

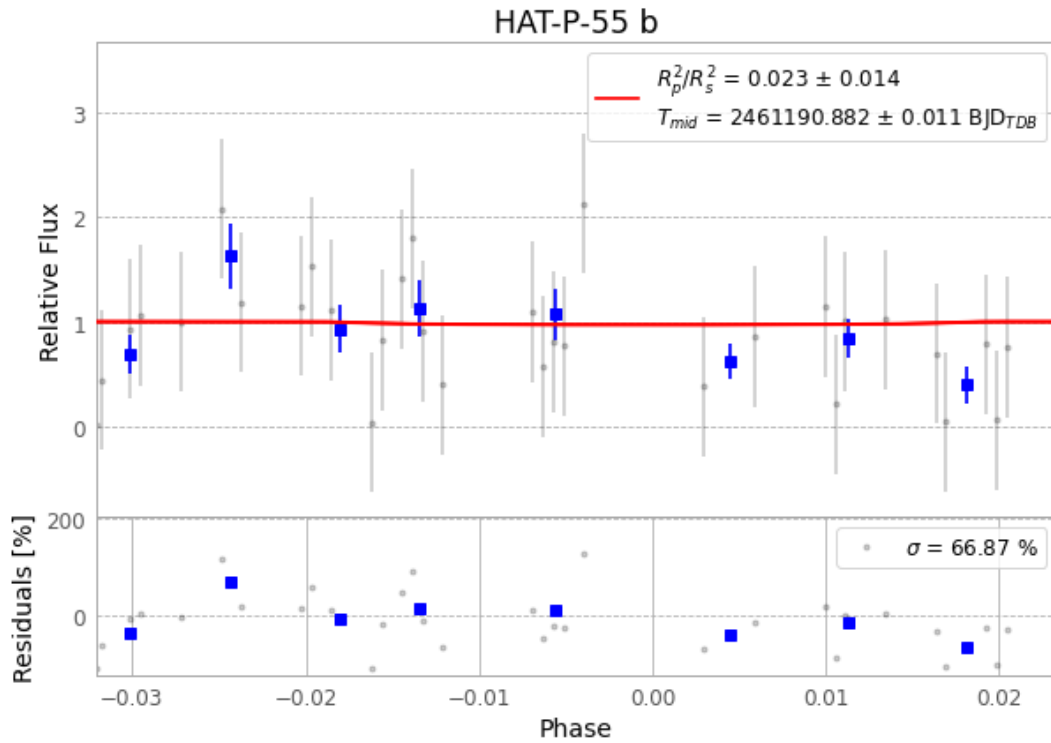


Figure 1 — FinalLightCurve_HAT-P-55 b_2026-05-30.png (SHA-256 10dbaebf3828782d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ee177568dcb21ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

97 FITS frames (64 MB), HATP-55260530061701.FITS → HATP-55260530110616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-260530.log (a0a6bc0b5258...), FinalLightCurve_HAT-P-55 b_2026-05-30.png (10dbaebf3828...), FinalLightCurve_HAT-P-55 b_2026-05-30.csv (a75a89024587...)

Compute resources

Wall time 179.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)